



Build Instructions - GoalPulse

Prerequisites

1. **Node.js 18+** or **Node.js 20+**

```
bash
node --version
```

2. **Yarn 4.x** (included in project)

```
bash
yarn --version
```

3. **EAS CLI** (Expo Application Services)

```
bash
npm install -g eas-cli
eas --version # Should be 13.2.0+
```

4. **Expo Account** (free)

- Sign up at: <https://expo.dev/signup>



Step-by-Step Build Process

Step 1: Install Dependencies

```
cd react_native_space
yarn install
```

Expected output:

```
✅ Done with warnings in ~10s
```

Step 2: Verify Setup (Optional but Recommended)

```
npx expo-doctor
```

Expected output:

```
✅ Running 17 checks on your project...
✅ 17/17 checks passed. No issues detected!
```

Step 3: Login to Expo

```
npx eas login
```

Enter your Expo credentials.

Step 4: Configure EAS Build (First Time Only)

```
npx eas build:configure
```

This will:

1. Create/update `eas.json` (already included)
2. Link your project to your Expo account
3. Generate a project ID

You'll be asked:

- "What would you like your Android application id to be?"
- Press Enter to use default: `com.goalpulse.app`

Step 5: Build Android APK (Preview)

```
npx eas build --profile preview --platform android
```

Build process:

1. 📦 Uploads your project to EAS servers
2. 🔨 Builds APK on cloud (takes ~5-10 minutes)
3. 📱 Provides download link when complete

Expected output:

- ✓ Build started, it may take a few minutes to complete.
- ✓ You can monitor it at <https://expo.dev/accounts/.../builds/...>

When complete:

- ✓ Build finished!
- ✓ APK: <https://expo.dev/artifacts/.../build-123.apk>

Step 6: Download & Install APK

1. Click the APK link from EAS output
2. Transfer to Android device
3. Install the APK (enable "Install from Unknown Sources" if needed)

Build Profiles

Preview Profile (APK)

Use for: Testing on physical devices

```
npx eas build --profile preview --platform android
```

- Builds APK (can be directly installed)
- Internal distribution

- Faster builds (~5-10 min)

Production Profile (AAB)

Use for: Google Play Store submission

```
npx eas build --profile production --platform android
```

- Builds AAB (Android App Bundle)
- Required for Play Store
- Optimized for production

Local Development (Optional)

Run on Web (Quick Testing)

```
npx expo start --web
```

Run on Android Emulator

```
npx expo start --android
```

Requires Android Studio and emulator setup

Run on iOS Simulator (macOS only)

```
npx expo start --ios
```

Requires Xcode

Test with Expo Go App

```
npx expo start
```

Scan QR code with Expo Go app (iOS/Android)

 **Note:** Some features (notifications, camera) may not work in Expo Go. Use standalone build for full testing.

iOS Build (Optional)

Requirements

1. **Apple Developer Account** (\$99/year)
2. **macOS computer** (for local builds) OR use EAS Build (cloud)

Build iOS with EAS (Recommended)

```
npx eas build --profile preview --platform ios
```

Troubleshooting

“Command not found: eas”

```
npm install -g eas-cli
```

“expo-doctor fails”

```
npx expo install --fix
```

“Build fails on EAS”

1. Check build logs: `npx eas build:list`
2. Click on failed build to see detailed logs
3. Common issues:
 - Gradle timeout: Retry build
 - Dependency mismatch: Run `npx expo install --fix`

“Cannot install APK”

- Enable “Install from Unknown Sources” in Android settings
- Or: Settings → Security → Unknown Sources → Enable

“Notifications don’t work”

- Notifications only work in **standalone builds** (not Expo Go)
- Ensure you built with `eas build`, not running with `expo start`

What Gets Built

APK Contents

-  All app screens and features
-  Embedded SQLite database
-  Camera access for receipt scanning
-  Notification support
-  All assets (icons, images)
-  Optimized JavaScript bundle

APK Size

~40-60 MB (typical for Expo apps)

Next Steps After Build

1. **Test thoroughly** on real devices
 2. **Collect feedback** from users
 3. **Fix bugs** if found
 4. **Build production AAB** for Play Store
 5. **Submit to Google Play Console**
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Additional Resources

- **EAS Build Docs:** <https://docs.expo.dev/build/introduction/>
 - **Expo CLI Docs:** <https://docs.expo.dev/workflow/expo-cli/>
 - **Troubleshooting:** <https://docs.expo.dev/build-reference/troubleshooting/>
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Good luck with your build! 🎉